

Module 5: Digital Modulation



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Objective M5L30

- Threshold Detection
- Matched Filter



Threshold Detection

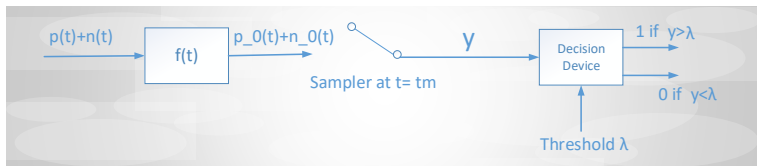
Leibnitz's Rule:

$$\frac{d}{dx} \left(\int_{a(x)}^{b(x)} f(x, t) dt \right) = f(x, b(x)) \cdot \frac{d}{dx} b(x) - f(x, a(x)) \cdot \frac{d}{dx} a(x) + \int_{a(x)}^{b(x)} \frac{\partial}{\partial x} f(x, t) dt.$$

Threshold Detection for Polar with Unequal Probability

$$\lambda = \frac{\sigma^2}{2A_p} \ln \frac{p_0}{p_1} \quad (1)$$

Receiver: $f(t)$



- Determine the filter $f(t)$ that minimizes the BER.
- $y(t_m) = \pm p_0(t_m) + n_0(t_m)$.
- What is BER?
- $P_e = Q\left(\frac{A_p}{\sigma_n}\right)$
- $P_e = Q\left(\sqrt{\frac{A_p^2}{\sigma_n^2}}\right)$
- $P_e = Q(\sqrt{SNR}) = Q(\sqrt{\gamma})$
- γ : SNR
- Minimizing the BER is same as maximizing the SNR.

Objective M5L31

- Matched Filter
- General Transmission



Receiver: $f(t)$ (Minimum BER Criterion)

- $\gamma = \frac{p_0^2(t_m)}{\sigma_n^2}$
- $p_0(t)$?
- $p_0(t) = \int_{-\infty}^{\infty} P(f)F(f)e^{j2\pi ft}df$
- $p_0(t_m) = \int_{-\infty}^{\infty} P(f)F(f)e^{j2\pi ft_m}df$
- $\sigma_n^2 = \int_{-\infty}^{\infty} S_n(f)|F(f)|^2df$
- $\gamma = \frac{|\int_{-\infty}^{\infty} P(f)F(f)e^{j2\pi ft_m}df|^2}{\int_{-\infty}^{\infty} S_n(f)|F(f)|^2df}$

Receiver Structure: Matched Receiver

- Cauchy-Schwarz inequality:
$$|\int_{-\infty}^{\infty} \phi_1(x)\phi_2(x)dx|^2 \leq \int_{-\infty}^{\infty} |\phi_1(x)|^2 dx \int_{-\infty}^{\infty} |\phi_2(x)|^2 dx$$
- Equality $\phi_1(x) = k\phi_2^*(x)$
- $\phi_1(x) = F(f)\sqrt{S_n(f)}$, $\phi_2(x) = \frac{P(f)e^{j2\pi f t_m}}{\sqrt{S_n(f)}}$
- Apply Cauchy-Schwarz inequality:
- $\gamma \leq \int_{-\infty}^{\infty} \frac{|P(f)|^2}{S_n(f)} df$
- Maximized if $F(f)\sqrt{S_n(f)} = k[\frac{P(f)e^{j2\pi f t_m}}{\sqrt{S_n(f)}}]^*$
- $F(f) = k\frac{P(-f)e^{-j2\pi f t_m}}{S_n(f)}$
- Matched to pulse !

Matched Filtering: Condition

- Filtering = Receiver Structure
- $F(f) = k \frac{P(-f)e^{-j2\pi f t_m}}{S_n(f)}$
- Depends on: pulse shape, sampling instant, noise PSD

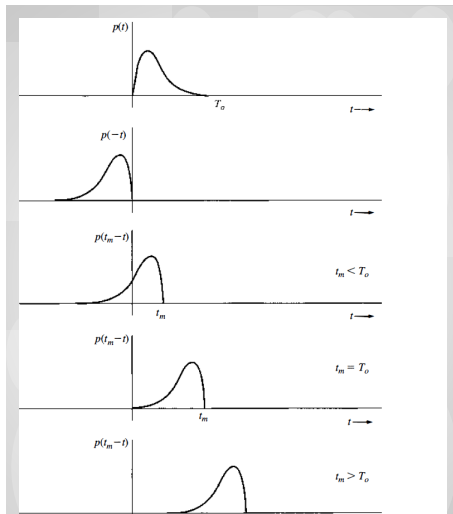
Matched Filtering: Gaussian Noise

- $F(f) = k \frac{P(-f)e^{-j2\pi f t_m}}{S_n(f)}$
- Consider Gaussian noise $S_n(f) = N_0/2$
- $F(f) = \frac{2k}{N_0} P(-f)e^{-j2\pi f t_m}$
- $f(t) = \frac{2k}{N_0} p(t_m - t)$
- Matched to pulse
- $\gamma_{max} = \int_{-\infty}^{\infty} \frac{|P(f)|}{S_n(f)} df$
- $\gamma_{max} = \frac{2}{N_0} \int_{-\infty}^{\infty} |P(f)|^2 df$
- $\gamma_{max} = \frac{2E_P}{N_0}$
- $P_e = Q(\sqrt{\frac{2E_P}{N_0}})$
- Pulse energy: $E_p = \int_{-\infty}^{\infty} |P(f)|^2 df = \int_{-\infty}^{\infty} |p(t)|^2 dt$
- BER is independent of pulse shape !

Matched Filtering: Sampling Instant

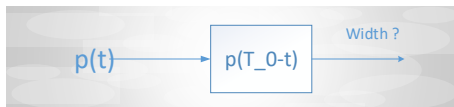
Pulse width = T_0 .

Which t_m is good?



Matched Filtering: Example

- If the width of $p(t)$ is T_0 , what is the width of MF $f(t)$?
- T_0
- What is the width at the output of MF?



- $2T_0$.

Matched Filtering: Example

Find the matched filter and output signal at the MF if the input signal pulse is a rectangular pulse of width T_0 . Find BER for polar with equi-probable transmission.

Matched Filter: On Transmit or Received?

CW Problem

Find the matched filter and output signal at the MF if the input signal pulse is a rectangular pulse of width 1. Channel is $\delta(t - 5)$.

Objective M5L32

- MF: Input and Output performance
- Matched Filter
- General Transmission

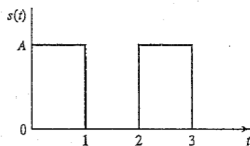


CW: MF Performance

Find SNR before and after the MF.

CW Problem

For binary polar signaling, the received signal is $y(t) = s(t) + n(t)$, where $n(t)$ is AWGN with PSD $N_0/2$ and $p(t)$ is as shown Figure 1.



- 1 Sketch the impulse response of the filter matched to $s(t)$.
- 2 Sketch the output of the matched filter matched to the input $s(t)$.
- 3 Determine the SNR at the output of MF.
- 4 Determine the probability of error.

Solution

$T_b = 3$, $p(t) = s(t)$, $q(t) = -s(t)$. Matched filter

$$f(t) = s(3 - t) = s(t)$$

Output of MF $r(t) = s(t) * f(t) + n(t) * f(t)$ Signal output
 $s(t) * f(t) = \int_{-\infty}^{\infty} s(t)f(t - \tau)d\tau$. Directly write convolution output:

$$r(t) = \begin{cases} 0, t < 0 \\ A^2 t, 0 \leq t < 1 \\ A^2(2 - t), 1 \leq t < 2 \\ 2A^2(t - 2), 2 \leq t < 3 \\ 2A^2(4 - t), 3 \leq t < 4 \\ A^2(t - 4), 4 \leq t < 5 \\ A^2(6 - t), 4 \leq t < 5 \\ 0, t \geq 6 \end{cases}$$

Solution

Signal power at the MF output at $t = T_b$

$r(T_b = 3) = 2A^2(4 - 3) = 2A^2$, Output signal power $|r(T_b)|^2 = 4A^4$.

Output noise power $P_n = \sigma_n^2 = \int_{-\infty}^{\infty} |F(f)|^2 S_n(f) df$. Output noise

power $P_n = \sigma_n^2 = \frac{N_0}{2} \int_0^3 |s(t)|^2 dt$. Output noise power

$P_n = \sigma_n^2 = \frac{N_0}{2} 2A^2 = N_0 A^2$. Output SNR $= \frac{4A^4}{N_0 A^2} = \frac{4A^2}{N_0}$.

Probability of error using output signals: General $P_e = Q(A_p/\sigma_n)$: where A_p and σ_n are signal parameters at the output of MF.

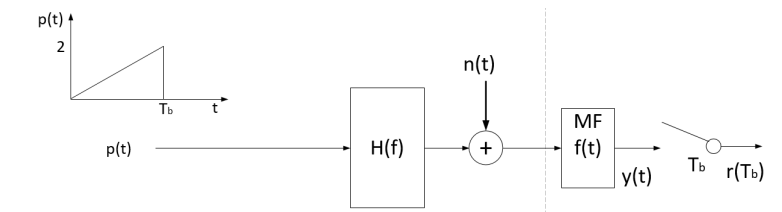
$P_e = P_b = Q(\sqrt{SNR}) = Q(\sqrt{\frac{4A^2}{N_0}})$. Probability of error using

input signals: using direct formula, polar: $P_e = Q(\sqrt{2E_p/N_0})$,

$E_p = \int_0^3 |s(t)|^2 dt = 2A^2$, polar: $P_e = Q(\sqrt{4A^2/N_0})$.

CW Problem

Let signal $p(t)$ is used as OOK modulation over channel $H(f)$ as shown in figure below. The double sided AWGN noise PSD is $N_0/2 = 4$ and the channel $H(f) = (1 + \cos 2\pi f T_b) e^{-j4\pi f}$. Find the MF $f(t)$ and the average BER in terms of Q-function.



Solution 3

$$\textcircled{0}. 2(a) \quad p(t) = \frac{2}{T_b} t$$

$$h(t) = \delta(t-2) + \frac{1}{2} \delta(t-2-T_b) + \frac{1}{2} \delta(t-2+T_b)$$

Signal before the MF

$$y(t) = p(t-2) + \frac{1}{2} p(t-2-T_b) + \frac{1}{2} p(t-2+T_b) \quad \textcircled{1}$$

$$\begin{aligned} \text{MF } f(t) &= p(T_b-t+2) + \frac{1}{2} p(T_b-t+2+T_b) \\ &\quad + \frac{1}{2} p(T_b-t+2-T_b) \\ &= p(T_b-t+2) + \frac{1}{2} p(2T_b-t+2) \\ &\quad + \frac{1}{2} p(-t+2) \quad \textcircled{1} \end{aligned}$$

$$\begin{aligned} \text{Energy of } y(t) &= \frac{4T_b}{3} + \frac{1}{4} \frac{4T_b}{3} + \frac{1}{4} \frac{4T_b}{3} \\ &= 2T_b \end{aligned}$$

$$P_e = Q \sqrt{\frac{E_p}{2N_0}} = Q \sqrt{\frac{T_b}{8}} \quad \textcircled{2}$$

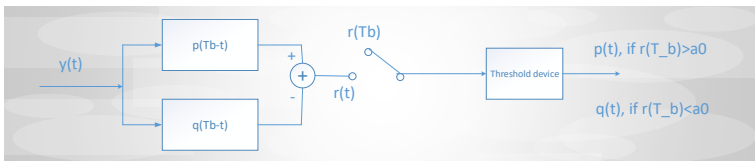
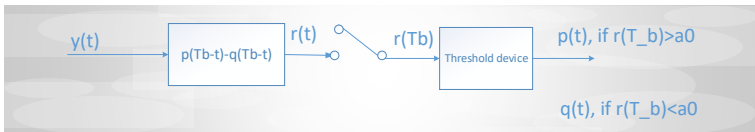
General Binary Signaling

- Previously, we restricted to polar line code.
- General situation
- $y(t) = \begin{cases} p(t) + n(t), 0 \leq t \leq T_b & \text{bit 1} \\ q(t) + n(t), 0 \leq t \leq T_b & \text{bit 0} \end{cases}$
- $r(T_b) = q(T_b) + n(T_b)$ for bit $m = 0$
- $r(T_b) = p(T_b) + n(T_b)$ for bit $m = 1$
- $m = \begin{cases} 0, r < a_0 \\ 1, r > a_0 \end{cases}$

General Binary Signaling: BER

- $P(\epsilon|0) = P(r > a_0) = Q\left(\frac{a_0 - q_0(T_b)}{\sigma_n}\right)$
- $P(\epsilon|1) = P(r < a_0) = Q\left(\frac{p_0(T_b) - a_0}{\sigma_n}\right)$
- Average BER $P_e = \frac{1}{2}Q\left(\frac{a_0 - q_0(T_b)}{\sigma_n}\right) + \frac{1}{2}Q\left(\frac{p_0(T_b) - a_0}{\sigma_n}\right)$
- Optimum threshold: derivative of P_e with respect to a to zero
- Show that optimal threshold: $a_0 = \frac{p_0(T_b) + q_0(T_b)}{2}$
- $f(t) = p(T_b - t) - q(T_b - t).$
- $E_{pq} = \int_0^{T_b} p(t)q(t)dt$
- BER $P_e = Q\left(\sqrt{\frac{E_p + E_q - 2E_{pq}}{2N_0}}\right)$

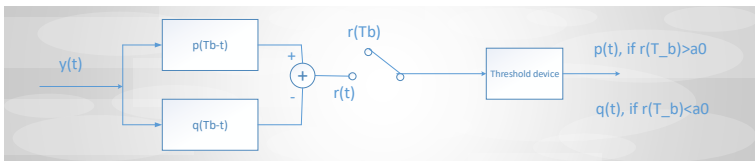
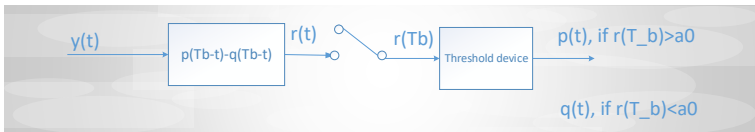
Receiver Structures



General Binary Signaling

- General situation
- $y(t) = \begin{cases} p(t) + n(t), 0 \leq t \leq T_b & \text{bit 1} \\ q(t) + n(t), 0 \leq t \leq T_b & \text{bit 0} \end{cases}$
- Show that optimal threshold: $a_0 = \frac{p_0(T_b) + q_0(T_b)}{2}$
- Or, $a_0 = \frac{E_p - E_q}{2}$
- $f(t) = p(T_b - t) - q(T_b - t)$.
- $E_{pq} = \int_0^{T_b} p(t)q(t)dt$
- BER $P_e = Q\left(\sqrt{\frac{E_p + E_q - 2E_{pq}}{2N_0}}\right)$

Receiver Structures



Special case: Polar Signaling

- Transmitted pulses: $p(t)$ for bit 1, and $q(t) = -p(t)$ for bit 0.
- $E_p = E_q$, and $E_{pq} = -E_p$
- BER $P_e = Q(\sqrt{\frac{E_p + E_q - 2E_{pq}}{2N_0}})$
- BER $P_e = Q(\sqrt{\frac{2E_p}{N_0}})$
- Average bit energy $E_b = (E_p + E_q)/2 = E_p$
- BER $P_b = Q(\sqrt{\frac{2E_b}{N_0}})$
- Threshold $a_0 = 0$.
- $f(t) = p(T_b - t) - q(T_b - t) = 2p(T_b - t)$
- Draw receiver structure

Special case: On-Off Signaling

- Transmitted pulses: $p(t)$ for bit 1, and $q(t) = 0$ for bit 0.
- $E_p, E_q = 0$, and $E_{pq} = 0$
- BER $P_e = Q(\sqrt{\frac{E_p + E_q - 2E_{pq}}{2N_0}})$
- BER $P_e = Q(\sqrt{\frac{E_p}{2N_0}})$
- Average bit energy $E_b = (E_p + E_q)/2 = E_p/2$
- BER $P_b = Q(\sqrt{\frac{E_b}{N_0}})$
- Threshold $a_0 = E_p/2$.
- $f(t) = p(T_b - t)$
- Compare on-off signaling with polar.

Special case: Orthogonal Signaling

- Select $p(t)$ and $q(t)$ such that $\int_0^{T_b} p(t)q(t)dt = 0$
- E_p , E_q , and $E_{pq} = 0$
- BER $P_e = Q(\sqrt{\frac{E_p + E_q - 2E_{pq}}{2N_0}})$
- BER $P_e = Q(\sqrt{\frac{E_p + E_q}{2N_0}})$
- Average bit energy $E_b = (E_p + E_q)/2$
- BER $P_b = Q(\sqrt{\frac{E_b}{N_0}})$
- Threshold $a_0 = (E_p - E_q)/2$.
- Compare on-off signaling, orthogonal, and polar.

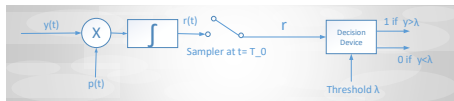
Objective M5L33

- General Transmission



Matched Filtering: Correlator Implementation

- Input pulse $y(t) = \pm p(t) + n(t)$
- Output at the MF $r(t) = \int_{-\infty}^{\infty} y(x)f(t-x)dx$
- Since $f(t) = p(T_0 - t)$
- $r(t) = \int_{-\infty}^{\infty} y(x)p(T_0 - (t-x))dx$
- $r(t) = \int_{-\infty}^{\infty} y(x)p(T_0 - t + x)dx$
- Sampled at $t = T_0$: $r(T_0) = \int_0^{T_0} y(x)p(x)dx$
- Correlator Receiver



Digital Passband Modulation: Coherent Vs Non-Coherent

- ASK, FSK, PSK
- Coherent: At the receiver, we know the phase and frequency of the carrier signal
- Non-coherent: we do not know the phase and frequency of the carrier signal, especially the phase.

Special case: Coherent BPSK

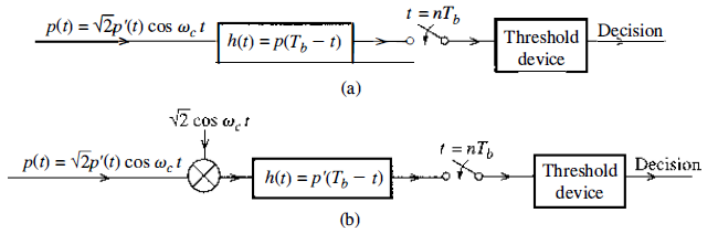
- BPSK modulation
- $p'(t)$: Baseband pulse shaping
- bit 1: $\sqrt{2}p'(t) \cos(2\pi f_c t)$, $0 \leq t \leq T_b$ bit 1
- bit 0: $-\sqrt{2}p'(t) \cos(2\pi f_c t)$, $0 \leq t \leq T_b$ bit 1
- Draw the transceiver structure for coherent BPSK
- The above scheme is exactly same as polar with
 $p(t) = \sqrt{2}p'(t) \cos(2\pi f_c t)$
- BER $P_e = Q(\sqrt{\frac{2E_p}{N_0}}) = Q(\sqrt{\frac{2E_b}{N_0}}) = P_b$
- $E_p = \int_0^{T_b} p^2(t) dt = 2 \int_0^{T_b} [p'(t)]^2 \cos^2(2\pi f_c t) dt = E'_p$

Special Case: Coherent BASK

- BASK modulation
- $p'(t)$: Baseband pulse shaping
- bit 1: $\sqrt{2}p'(t) \cos(2\pi f_c t)$
- bit 0: 0
- Draw the transceiver structure for coherent BASK
- The above scheme is exactly same as on-off with $p(t) = \sqrt{2}p'(t) \cos(2\pi f_c t)$
- BER $P_b = Q\left(\sqrt{\frac{E_b}{N_0}}\right)$
- $E_p = \int_0^{T_b} p^2(t) dt = 2 \int_0^{T_b} [p'(t)]^2 \cos^2(2\pi f_c t) dt = E'_p$
- $E_b = E_p/2 = E'_p/2$

Coherent Receiver Structures: BPSK, ASK

- Compare ASK and BPSK for BER performance



Special Case: Coherent BFSK

- $p(t) = \sqrt{2}p'(t) \cos(2\pi(f_c - \Delta f/2)t)$
- $q(t) = \sqrt{2}p'(t) \cos(2\pi(f_c + \Delta f/2)t)$

Coherent MSK

- $p(t) = \sqrt{2}p'(t) \cos(2\pi(f_c - \Delta f/2)t)$
- $q(t) = \sqrt{2}p'(t) \cos(2\pi(f_c - \Delta f/2)t)$
- If $\Delta f = \frac{1}{2T_b}$: MSK (minimum shift keying)

Basis, Constellation, and Signal Space Analysis

Objective M5L34

- Basis Signals
- Constellation

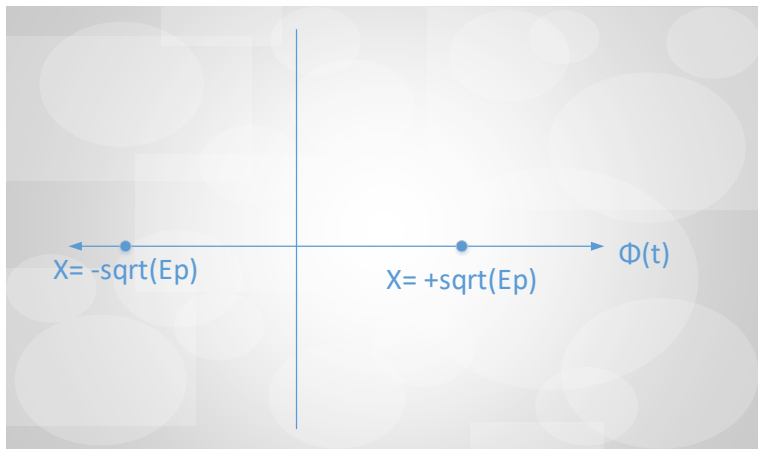


Constellation: Polar Signaling

- Polar: $p(t)$ and $-p(t)$
- Basis signal $\phi_1(t) = \frac{p(t)}{\sqrt{E_p}}$
- Basis signal $\phi_2(t) = 0$
- Thus $p(t) = \sqrt{E_p}\phi_1(t)$
- $-p(t) = -\sqrt{E_p}\phi_1(t)$
- In general: one dimensional signals can be represented by a single basis signal.
- Corresponding vectors ($\sqrt{E_p}$ and $-\sqrt{E_p}$) are called: constellation points or symbols.
- Distance between two constellation points $d = 2\sqrt{E_p}$
- Constellation points must either orthogonal $\rho = 0$ or highly uncorrelated $\rho = -1$.

Constellation: Polar Signaling

- Polar: $p(t)$ and $-p(t)$
- Distance between two constellation points $d = 2\sqrt{E_p}$



BER: Polar

- $y = x_1 + n = \sqrt{E_p} + n$
- $P(\epsilon|x_1) = p(\sqrt{E_p} + n < 0)$
- $P(\epsilon|x_1) = Q(\sqrt{\frac{2E_p}{N_0}})$
- Similarly:
- $P(\epsilon|x_0) = p(-\sqrt{E_p} + n > 0) = Q(\sqrt{\frac{2E_p}{N_0}})$

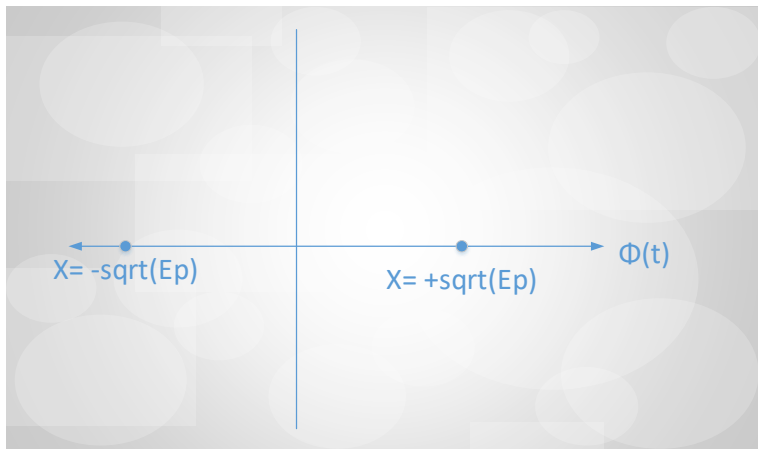
Basis, Constellation, and Signal Space Analysis

Constellation: Polar Signaling

- Polar: $p(t)$ and $-p(t)$
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- Basis signal $\phi_2(t) = 0$
- Thus $p(t) = \sqrt{E_p}\phi_1(t)$
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Constellation: Polar Signaling

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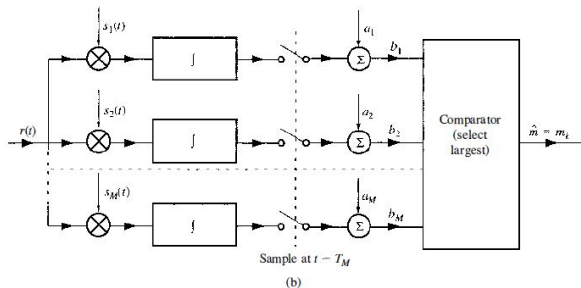
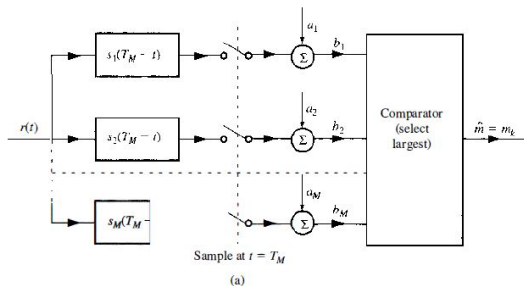
BER: Polar

- $y = x_1 + n = \sqrt{E_p} + n$
- $P(\epsilon|x_1) = p(\sqrt{E_p} + n < 0)$
- $P(\epsilon|x_1) = Q(\sqrt{\frac{2E_p}{N_0}})$
- Similarly:
- $P(\epsilon|x_0) = p(-\sqrt{E_p} + n > 0) = Q(\sqrt{\frac{2E_p}{N_0}})$

Receiver Structures: General M-ary

- Transmitted pulses $p_1(t), p_2(t), \dots, p_M(t)$ for M -symbols.
- Both transmitter and receiver should have knowledge of the M -waveforms/pulses.
- Receiver must apply M number of MF. Complex for higher M
- Criterion for M -waveforms: dissimilar or orthogonal.
- We should standardized the waveforms for digital transmission.
- If number of MF branches decreases: bonus
- Signal space analysis for digital transmission.

Receiver Structures



Objective M5L35

- SER for Digital Modulation Techniques

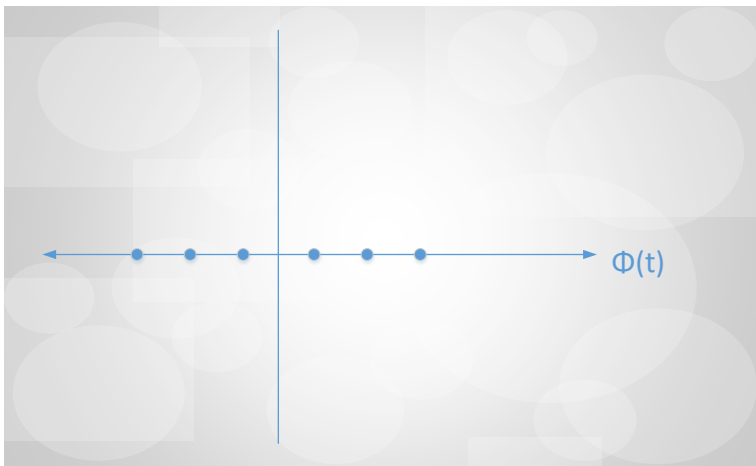


M-ary PAM

- Transmitted pulses $x_1(t), x_2(t), \dots, x_M(t)$ for M -symbols.
- Both transmitter and receiver should have knowledge of the M -waveforms/pulses.
- $\pm p(t), \pm 3p(t), \pm 5p(t), \dots + \pm(M-1)p(t)$,
- A single basis signal can represent one-dimensional signal.
- Basis vector $\phi_1(t) = \frac{p(t)}{\sqrt{E_p}}$
- Thus $\pm\sqrt{E_p}\phi(t), \pm 3\sqrt{E_p}\phi(t), \pm 5\sqrt{E_p}\phi(t), \dots + \pm(M-1)\sqrt{E_p}\phi(t)$

Constellation: M-ary PAM

- Distance between two constellation points $d = 2\sqrt{E_p}$



Pulse Energy, Symbol Energy, Bit Energy, Bit Power

- A point in constellation is just a symbol point and not a symbol.
- A baseband symbol is $x(t) = x\phi(t)$, where x is constellation point.
- A passband symbol is $x(t) = x\phi(t) \cos 2\pi f_c t$, where x is constellation point.
- Symbol energy = pulse energy i.e., $E_s = E_p$.
- Energy per bit: $E_b = \frac{E_s}{\log_2 M} = \frac{E_p}{\log_2 M}$.
- For binary $E_b = E_p = E_s$.
- Signal power: $P_s = E_s/T_s = E_s R_s$.
- Bit power $P_b = E_b/T_b = E_b R_b$

Average Energy of Pulse : M-ary PAM

- $\bar{E}_{pM} = \frac{2}{M}[E_p + 9E_p + 25E_p + \dots + (M-1)^2 E_p]$
- $\bar{E}_{pM} = E_p \frac{2}{M} \sum_{k=0}^{(M-2)/2} (2k+1)^2$
- $\bar{E}_{pM} = \frac{M^2-1}{3} E_p$
- For binary $M=2$: $\bar{E}_{pM} = E_{p2}$.
- Average energy increases by a factor $M^2/3$
- Bit rate transmissions increases by a factor $\log_2 M$
- Average energy per bit $E_b = \frac{\bar{E}_{pM}}{\log_2 M}$
- Average symbol power $\bar{P}_{pM} = \frac{\bar{E}_{pM}}{T_s} = \bar{E}_{pM} R_s$
- $E_p = \frac{3 \log_2 M}{M^2-1} E_b$

Objective M5L36

- SER for Digital Modulation Techniques



Constellation: Energy versus SER

M-ary Bandpass PSK

- $x(t) = p(t) \cos(2\pi f_c t + \frac{2\pi m}{M}), m = 0, 1, 2, \dots, M-1, 0 \leq t \leq T_s$
- $x(t) =$
 $p(t) \cos(2\pi f_c t) \cos(2\pi m/M) - p(t) \sin(2\pi f_c t) \sin(2\pi m/M)$
- $x(t) = \sqrt{\frac{E_p}{2}} \cos(2\pi m/M) \phi_1(t) + \sqrt{\frac{E_p}{2}} \sin(2\pi m/M) \phi_1(t)$
- $\phi_1(t) = \sqrt{\frac{2}{E_p}} p(t) \cos 2\pi f_c t$
- $\phi_2(t) = -\sqrt{\frac{2}{E_p}} p(t) \sin 2\pi f_c t$

M-ary Bandpass PSK

- $x(t) = \sqrt{\frac{E_p}{2}} \cos(2\pi m/M) \phi_1(t) + \sqrt{\frac{E_p}{2}} \sin(2\pi m/M) \phi_2(t)$
- Constellation points in two-dimensional plane.
- $\left(\sqrt{\frac{E_p}{2}} \cos(2\pi m/M), \sqrt{\frac{E_p}{2}} \sin(2\pi m/M) \right)$
- Points: $x_m = \sqrt{\frac{E_p}{2}} \cos(2\pi m/M)$, $y_m = \sqrt{\frac{E_p}{2}} \sin(2\pi m/M)$
- $x(t) = x_m \phi_1(t) + y_m \phi_2(t)$
- $x_m^2 + y_m^2 = \frac{E_p}{2}$: All signal points occur on a circle.
- Draw 2-ary (BPSK), 4-ary (QPSK), and 8-ary PSK constellations.

M-ary QAM

- M-ary PAM: Amplitude modulation, and M-ary PSK: Phase modulation
- M-ary QAM: Combine amplitude and phase modulation.
- In M-ary PSK, the signal constellations lie on a circle such that each symbol has same amplitude i.e. equal energy.
- M-ary QAM: If M-ary PSK symbols have unequal amplitude i.e., unequal energy.

M-ary QAM

- In general: $x(t) = A_m p(t) \cos(2\pi f_c t + \theta_n)$,
 $m = 0, 1, 2, \dots, M_1 - 1$, $n = 0, 1, 2, \dots, M_2 - 1$, $0 \leq t \leq T_s$
- $x(t) = A_{mI} p(t) \cos(2\pi f_c t) + A_{mQ} p(t) \sin(2\pi f_c t)$,
 $m = 0, 1, 2, \dots, M - 1$
- $\{A_{mI}\}$ and $\{A_{mQ}\}$ are the sets of amplitude levels obtained by mapping $\log_2 M$ bits into signal amplitudes.

M-ary QAM

- $x(t) = A_{mI}p(t) \cos(2\pi f_c t) + A_{mQ}p(t) \sin(2\pi f_c t),$
 $m = 0, 1, 2, \dots M - 1$
- $\phi_1(t) = \sqrt{\frac{2}{E_p}}p(t) \cos 2\pi f_c t$
- $\phi_2(t) = -\sqrt{\frac{2}{E_p}}p(t) \sin 2\pi f_c t$
- $x(t) = \sqrt{\frac{E_p}{2}}A_{mI}\phi_1(t) + \sqrt{\frac{E_p}{2}}A_{mQ}\phi_2(t)$
- $x(t) = x_m\phi_1(t) + y_m\phi_2(t)$
- Constellation $\left(\sqrt{\frac{E_p}{2}}A_{mI}, \sqrt{\frac{E_p}{2}}A_{mQ} \right)$
- Points: $x_m = \sqrt{\frac{E_p}{2}}A_{mI}, y_m = \sqrt{\frac{E_p}{2}}A_{mQ}$
- $x_m^2 + y_m^2 = \text{not constant}$

Objective M5L37

- SER for Digital Modulation Techniques
- General Receiver Design



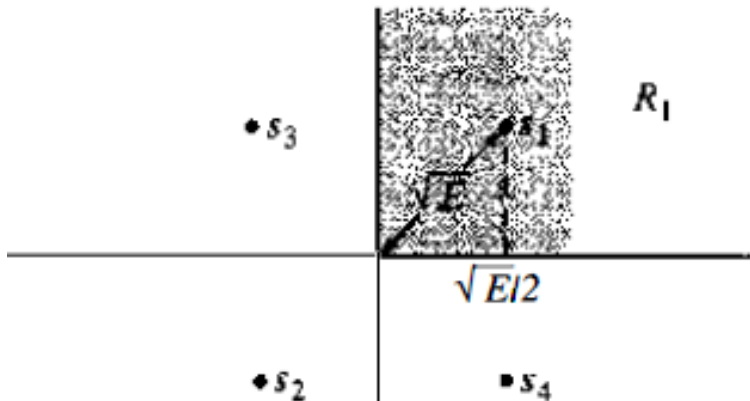
Average SER : M-ary PAM

- Extreme points = only a single neighbor i.e. binary case
- Other symbols: two neighbors
- $P_{eM} = \frac{1}{M} [Q(\sqrt{2E_p/N_0}) + Q(\sqrt{2E_p/N_0}) + (M-2)2Q(\sqrt{2E_p/N_0})]$
- $P_{eM} = 2(\frac{M-1}{M})Q(\sqrt{2E_p/N_0})$
- Since $E_p = \frac{3 \log_2 M}{M^2-1} E_b$
- $P_{eM} = 2(\frac{M-1}{M})Q(\sqrt{\frac{6 \log_2 M}{M^2-1} \frac{E_b}{N_0}})$

QPSK Constellation

Draw constellation

- $s_1 = -s_2 = \sqrt{E/2}\phi_1(t)$
- $s_2 = -s_4 = \sqrt{E/2}\phi_2(t)$
- Draw



- Correct probability:
- $P(C|m_1) = P(n_1 > -\sqrt{\frac{E}{2}}, n_2 > -\sqrt{\frac{E}{2}})$
- $P(C|m_1) = P(n_1 > -\sqrt{\frac{E}{2}})P(n_2 > -\sqrt{\frac{E}{2}})$
- $P(C|m_1) = [1 - Q(\sqrt{\frac{E}{2\sigma_n^2}})]^2$
- $P(C|m_1) = [1 - Q(\sqrt{\frac{E}{N_0}})]^2$
- $P(C|m_1) = [1 - Q(\sqrt{\frac{2E_b}{N_0}})]^2$
- Average correct probability: $P(C) = [1 - Q(\sqrt{\frac{2E_b}{N_0}})]^2$
- SER: $1 - P(C)$

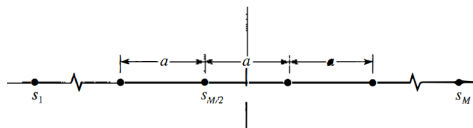
SER Derivation: General Approach

Use of distance between constellation points and expressing the distance in terms of $E_b = \frac{E}{\log_2 M}$

- All binary
- QPSK
- 8-ary QAM
- 16-ary QAM

Problem

A source emits M equiprobable messages, which are assigned signals $s_1, s_2, \dots, s_M$ as shown in Fig. Determine the optimum receiver and the corresponding error probability P_{eM} for an AWGN channel as a function of E_b/N .



Solution: SER

- Correct decision:

$$P(C|m_1) = P(n_I < \frac{a}{2}) = 1 - Q(\frac{a}{2\sigma_n}) = 1 - p.$$

- Correct decision:

$$P(C|m_M) = P(n_I > \frac{-a}{2}) = Q(\frac{-a}{2\sigma}) = 1 - Q(\frac{a}{2\sigma_n}) = 1 - p.$$

- Other Symbols $P(C|m_j)$, where $j = 2, 3, \dots, M-1 =$

$$P(|n_I| > \frac{a}{2}) = 1 - 2Q(\frac{a}{2\sigma}) = 1 - 2p: \text{ Show this}$$

- $P_{eM} = \frac{2(M-1)}{M} Q(\frac{a}{2\sigma_n})$

Solution: SER

Hence the average pulse energy $\overline{E}$ is

$$\begin{aligned}\overline{E} &= \frac{2}{M} \left[\frac{a^2}{4} + \frac{9a^2}{4} + \frac{25a^2}{4} + \dots + \frac{(M-1)^2 a^2}{4} \right] \\ &= \frac{a^2}{2M} \sum_{k=0}^{\frac{M-2}{2}} (2k+1)^2 = \frac{(M^2-1)}{12} a^2\end{aligned}$$

Also

$$E_b = \frac{\overline{E}}{\log_2 M} = \frac{(M^2-1)a^2}{12 \log_2 M}$$

SER for QPSK

SER for 8-QAM

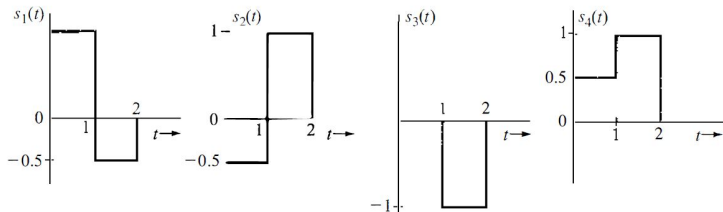
SER for 16-QAM

$$P_e = ?$$

$$P_e \approx 3Q\left(\sqrt{\frac{4E_b}{5N_0}}\right)$$

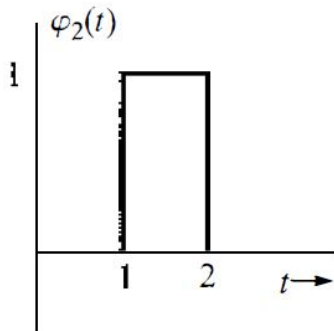
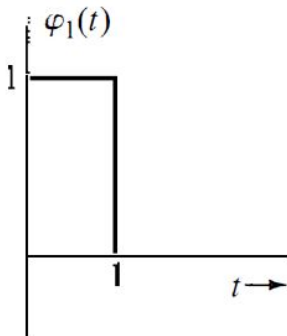
Signal Space Representation: Example 1

Find basis signals and signal space representation of the following:



Solution

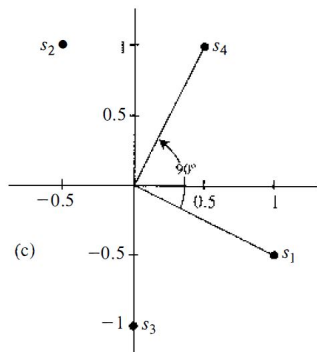
Basis signal (by inspection)



Solution

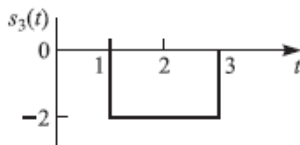
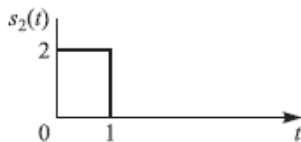
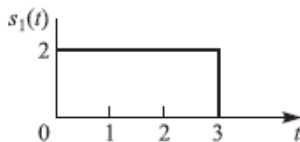
Signal representation:

- $\mathbf{s}_1 = (1, -0.5)$
- $\mathbf{s}_2 = (-0.5, 1)$
- $\mathbf{s}_3 = (0, -1)$
- $\mathbf{s}_4 = (0.5, 1)$



Signal Space Representation: Example 2

Find basis signals and signal space representation of the following:



Orthogonalization procedure: Gram Schmidt Process

- Given $x_1(t), x_2(t), \dots, x_M(t)$, find orthonormal basis signals.
- $\phi_1(t) = \frac{x_1(t)}{\sqrt{E_1}}$
- $E_1 = \int_{-\infty}^{\infty} |s_1(t)|^2 dt$
- Projection of $x_2(t)$ onto $\phi_1(t)$
- $c_{21} = \int_{-\infty}^{\infty} x_2(t)\phi_1(t)$
- Subtract $c_{21}\phi_1(t)$ from $x_2(t)$
- $d_2 = x_2(t) - c_{21}\phi_1(t)$
- $\phi_2(t) = \frac{d_2(t)}{\sqrt{E_2}}$
- $E_2 = \int_{-\infty}^{\infty} |d_2(t)|^2 dt$
- And so on....

GS Example 1

9.4-2 Given three linearly independent signals

$$s_1(t) = \begin{cases} 1 & \text{for } 0 \leq t \leq 1 \\ -1 & \text{for } 1 < t \leq 2 \\ 0 & \text{otherwise} \end{cases}$$

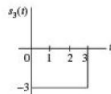
$$s_2(t) = \begin{cases} 2 & \text{for } 0 \leq t \leq 2 \\ 0 & \text{otherwise} \end{cases}$$

$$s_3(t) = \begin{cases} -3 & \text{for } 0 \leq t \leq 3 \\ 0 & \text{otherwise} \end{cases}$$

Obtain orthonormal basis function set using Gram-Schmidt procedure (Appendix C). Express $s_1(t)$, $s_2(t)$ and $s_3(t)$ in terms of basis functions.

Solution

The function $s_i(t)$, $i = 1, 2, 3$ are



Let $\phi_i(t)$ be the basis functions. Using Gram-Schmidt procedure

$$\phi_1(t) = \frac{s_1(t)}{\|s_1\|}$$

$$\|s_1\|^2 = \int_{-\infty}^{\infty} s_1^2(t) dt = 2$$

Hence,

$$\phi_1(t) = \frac{s_1(t)}{\sqrt{2}}$$

$$c_{12} = \int_{-\infty}^{\infty} s_2(t) \phi_1(t) dt = \int_0^1 2 \frac{1}{\sqrt{2}} dt + \int_1^2 2 \left(-\frac{1}{\sqrt{2}} \right) dt = 0$$

and

$$g_2(t) = s_2(t) - c_{12} \phi_1(t) = s_2(t)$$

and

$$\begin{aligned} \phi_2(t) &= \frac{g_2(t)}{\|g_2(t)\|} \\ &= \frac{s_2(t)}{\|s_2(t)\|} = \frac{s_2(t)}{\left[\int_{-\infty}^{\infty} s_2^2(t) dt \right]^{1/2}} \\ &= \frac{s_2(t)}{\sqrt{8}} \end{aligned}$$

$$g_3(t) = s_3(t) - c_{13} \phi_1(t) - c_{23} \phi_2(t)$$

Solution

$$c_{13} = \int_{-\infty}^{\infty} s_3(t)\phi_1(t)dt = \int_0^1 (-3)\left(\frac{1}{\sqrt{2}}\right)dt + \int_1^2 (-3)\left(-\frac{1}{\sqrt{2}}\right)dt$$

$$= 0$$

$$c_{23} = \int_{-\infty}^{\infty} s_3(t)\phi_2(t)dt$$

$$= \int_0^2 (-3)\left(\frac{2}{\sqrt{8}}\right)dt = \frac{12}{\sqrt{8}}$$

Hence,

$$g_3(t) = s_3(t) - \left(\frac{-12}{\sqrt{8}}\right)\phi_2(t)$$

$$= s_3(t) + \frac{12}{\sqrt{8}}\phi_2(t)$$

Nothing that

$$\phi_2(t) = \frac{s_2(t)}{\sqrt{8}} = \begin{cases} \frac{2}{\sqrt{8}} & \text{for } 0 \leq t \leq 2 \\ 0 & \text{otherwise} \end{cases}$$

and using $s_3(t)$ and $\phi_2(t)$ in the expression for $g_3(t)$,

$$g_3(t) = \begin{cases} -3 & \text{for } 2 \leq t \leq 3 \\ 0 & \text{otherwise} \end{cases}$$

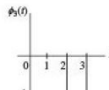
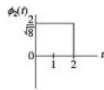
and

$$\|g_3(t)\|^2 = \int_2^3 g^2(t)dt = 9$$

Therefore,

$$\phi_3(t) = \frac{g_3(t)}{\|g_3(t)\|} = \begin{cases} -1 & \text{for } 2 \leq t \leq 3 \\ 0 & \text{otherwise} \end{cases}$$

$\phi_1(t)$, $\phi_2(t)$ and $\phi_3(t)$ are sketched below



Waveform and Vector AWGN Channels: Equivalence

- $r(t) = s_m(t) + n(t)$, $1 \leq m \leq M$
- Vector representation: $\mathbf{r} = \mathbf{s}_m + \mathbf{n}$
- All vectors are of size N : orthogonal expansion using N basis signals.
- $s_m(t) = \sum_{j=1}^N s_{mj} \phi_j(t)$
- $n_t(t) = \sum_{j=1}^N n_j \phi_j(t)$
- Let $n_j \sim (0, N_0/2)$
- What is PDF noise vector $\mathbf{n}$?
- $f(\mathbf{n}) = \left(\frac{1}{\sqrt{\pi N_0}}\right)^N e^{-\|\mathbf{n}\|^2/N_0}$

Optimum Detector Rule: ML

- $f[\mathbf{r}|\mathbf{s}_m]$: likelihood
- If $P_m = 1/M$ (equi-probable)
- $\hat{m} = \arg \max_{1 \leq m \leq M} f(\mathbf{r}|\mathbf{s}_m)$
- Maximum Likelihood receiver.

Optimum Detector for AWGN: ML

- $\hat{m} = \arg \max_{1 \leq m \leq M} [P_m f[\mathbf{r} | \mathbf{s}_m]]$
- $\hat{m} = \arg \max_{1 \leq m \leq M} [P_m f_{\mathbf{n}}(\mathbf{r} - \mathbf{s}_m)]$
- $\hat{m} = \arg \max_{1 \leq m \leq M} [P_m (\frac{1}{\sqrt{\pi N_0}})^N e^{-\|\mathbf{r} - \mathbf{s}_m\|^2 / N_0}]$
- $\hat{m} = \arg \max_{1 \leq m \leq M} [P_m e^{-\|\mathbf{r} - \mathbf{s}_m\|^2 / N_0}]$
- $\hat{m} = \arg \max_{1 \leq m \leq M} [\ln P_m - \frac{\|\mathbf{r} - \mathbf{s}_m\|^2}{N_0}]$
- $\hat{m} = \arg \max_{1 \leq m \leq M} [-\|\mathbf{r} - \mathbf{s}_m\|^2]$
- $\hat{m} = \operatorname{argmin}_{1 \leq m \leq M} [\|\mathbf{r} - \mathbf{s}_m\|^2]$
- ML receiver
- Nearest-neighbour or Minimum distance receiver or One shot receiver.
- The receiver receives $\mathbf{r}$ and looks among all $\mathbf{s}_m$ to find the one that is closest to $\mathbf{r}$ using standard Euclidean distance.
- Equal-probable and AWGN: MAP= ML

SER for M-PSK

- $D_1 = \{\theta : -\pi/M < \theta \leq \pi/M\}$
- $P_e = 1 - \int_{-\pi/M}^{\pi/M} f(\theta) d\theta$
- $M = 2, P_e = Q(\sqrt{\frac{2E_b}{N_0}})$
- $M = 4, P_e = 2Q(\sqrt{\frac{2E_b}{N_0}})[1 - \frac{1}{2}Q(\sqrt{\frac{2E_b}{N_0}})]$

SER for M-ary Orthogonal Modulation

- $\mathbf{s}_m = (\sqrt{E}, 0, \dots, 0), \dots, (0, 0, \dots, \sqrt{E})$
- $\mathbf{r} = (\sqrt{E} + n_1, n_2, n_3, \dots, n_M)$
- Distance between two points: $\sqrt{2E}$
- Detector $R_m = \mathbf{r} \cdot \mathbf{s}_m$
- Assume s_1 is transmitted.
- $R_1 = E + \sqrt{E}n_1, R_m = \sqrt{E}n_m, 2 \leq m \leq M$
- Suppose $\mathbf{s}_1$ was transmitted.
- Correct $P_c = \text{Prob}[R_1 > R_2, R_1 > R_3, \dots, R_1 > R_m | \mathbf{s}_1]$
- $P_c = P[\sqrt{E} + n_1 > n_2, \sqrt{E} + n_1 > n_3, \dots, \sqrt{E} + n_1 > n_m | \mathbf{s}_1]$
- Common term n_1 in all the events: not-independent
- $P_c = \int_{-\infty}^{\infty} P[\sqrt{E} + n > n_2, \sqrt{E} + n > n_3, \dots, \sqrt{E} + n > n_m | \mathbf{s}_1, n_1 = n] f_{n1}(n) dn$
- $P_c = \int_{-\infty}^{\infty} P[\sqrt{E} + n > n_m, | \mathbf{s}_1, n_1 = n]^{M-1} f_{n1}(n) dn$